Product Fusion Based Moving Target Detection Metric with Distributed SAR Systems

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***Abstract*—Distributed synthetic aperture radar (SAR) systems receive diverse electromagnetic signals from different angles, presenting great application potentials in ground moving target indication (GMTI). This paper proposes a moving target detection metric that employs a product rule to fuse both the magnitude and phase information in distributed multichannel SAR systems. For a distributed radar network with L M-channel SARs, this method first formulates a local detection test in each SAR by exploiting the magnitude of M-channel clutter-suppression residual and the interferometric phase between two clutter-suppression residuals computed from the first and last $M-1$ channels, respectively. Based on these estimation results, a global detection test is developed by fusing the local tests. As leveraging the multi-angle information can address the blind-speed detection problem occurred in the single SAR-GMTI system, the proposed detector achieves enhanced detection performance for slow and weak targets. The receiver operator characteristic metrics predicted through the Monte Carlo simulation method verify the effectiveness of the proposed method in the detection of dim moving targets.**

***Index Terms*—Distributed radar, weak moving target detection, product-based fusion.**

I. Introduction

Due to high-speed motions of airborne synthetic aperture radar (SAR) platforms, Doppler spectral of ground clutter is significantly broadening, leading to slowly-moving targets to be covered and thereby undetectable from background during ground moving target indication (GMTI). Multichannel SAR systems typically employ clutter-suppression methods, such as space-time adaptive processing (STAP) [1], to suppress stationary clutter and detect weak targets from clutter backgrounds [2]–[4]. However, in heterogeneous environments [5], as independent and identically distributed (i.i.d.) training samples are limited, the estimation accuracy of the clutter-plus-noise covariance matrix (CCM) may be degraded, resulting in extensive false alarms caused by isolated and strong clutter residuals in moving target detection. Recently, plenty of methods, that combine both magnitude features and the interferometric phase across multichannel SAR systems, have obtained detection performance improvements in heterogeneous backgrounds [6]–[8]. Nevertheless, from the perspective of single radar, moving targets may present slow radar-line-of-sight (radial) velocities and weak backscatter signals, thereby limiting target detection performance in these single-radar-based detection methods.

Distributed radar networks can observe moving targets from different angles to capture diverse motions and radar cross section (RCS) responses of moving targets, thereby providing more useful information for the detection of slow and weak targets [9], [10]. In [11], the exponentially weighted method based on the signal-to-noise ratio (SNR), which is referred to as the SNR-Exponentially Weighted (SEW) method, employs soft decision via the fusion of local statistics in each radar node to integrate all moving target features in distributed radar networks, which effectively enhance the signal-to-clutter-plus-noise ratio (SCNR), and thereby improve target detection performance. However, the SEW method only accumulates signal power from each radar, which may be susceptible to strong clutter and cause false alarms in heterogeneous background. The optimal fusion framework in [12] formulates local test in each multichannel radar with the full spatial DoFs to suppress heterogeneous clutter, and then, optimally fusing the local decisions for a global detection, significantly improving low-SCNR target detection performance in complex backgrounds. Nevertheless, as the optimal fusion strategy in this method needs prior information of targets to determine the fusing weight, it may suffer from performance degradation in practice when the target prior information is unknown or inaccurate.

To address this, this paper proposes a moving target detection method based on product fusion with distributed

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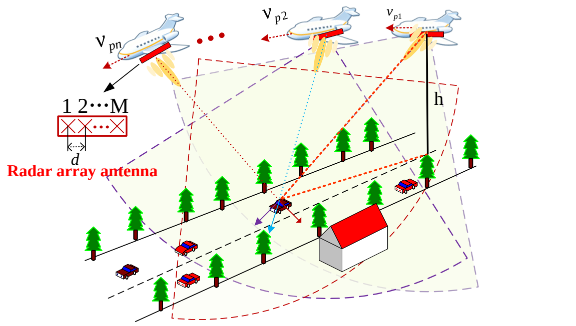

Fig. 1. Geometric relationship between the ground target and distributed radar system.

multichannel SAR systems. This method first formulates a local detection test by combining the residual magnitude and phase factor [13] (hereinafter referred to as the γ-factor detector) to enhance local clutter suppression, and then applies a product rule to construct a global detection test for final decision. Through several simulations, the results preliminarily demonstrate the effectiveness of the proposed method in the detection of slow and weak targets in coping with complex observation environments.

II. Signal Model Of Distributed Radar System

Consider a distributed radar network comprising L spatially separated airborne SAR nodes, each equipped with an M-channel uniform linear array aligned along the flight track with spacing d, as shown in Fig.1. The platforms move with velocities $v_{\mathrm{p}l}$ $(l = 1, \ldots, L)$ in a side-looking mode. Assuming orthogonal waveform transmission and independent reception, M co-registered SAR images are generated for each node following standard imaging and calibration processes, with all images spatially aligned to the reference node.

For a pixel k, let $\mathbf{z}_l(k) = [z_{l,1}(k), z_{l,2}(k), \ldots, z_{l,M}(k)]^{\mathrm{T}}$ denote the snapshot vector from the l-th node, where $[\cdot]^T$ represents the transposition and $z_{l,m}(k)$ represents the complex signal in the m-th SAR image of the l-th radar node. For ground moving target detection, a binary hypothesis test is defined as

$$\begin{cases} H_0 : \mathbf{z}_l(k) = \mathbf{c}_l(k) + \mathbf{n}_l(k) \\ H_1 : \mathbf{z}_l(k) = \mathbf{s}_l(k) + \mathbf{c}_l(k) + \mathbf{n}_l(k) \end{cases} \tag{1}$$

where H_0 and H_1 denote the target-absent and target-present hypotheses respectively. The vectors $\mathbf{s}_l(k)$, $\mathbf{c}_l(k)$, and $\mathbf{n}_l(k)$ represent the target, clutter, and noise components of the l-th radar. The noise vector is modeled as a zero-mean complex Gaussian vector with power $\sigma_{\mathrm{n}l}^2$, i.e., $\mathbf{n}_l(k) \sim \mathcal{CN}(\mathbf{0}, \sigma_{\mathrm{n}l}^2 \mathbf{I}_M)$.

In the l-th node, the target signal with radial velocity $v_{\mathrm{r}l}$ is modeled as $\mathbf{s}_l(k) = x_{\mathrm{t}l}(k)\mathbf{a}_{\mathrm{t}l}(k)$, where $x_{\mathrm{t}l}$ is the complex amplitude. Its spatial steering vector $\mathbf{a}_{\mathrm{t}l}(k)$ is given by:

$$\mathbf{a}_{\mathrm{t}l}(k) = \left[1, \ldots, \exp\left(j \frac{2\pi(M-1)dv_{\mathrm{r}l}(k)}{\lambda v_{\mathrm{p}l}}\right)\right]^{\mathrm{T}}, \tag{2}$$

where λ is the radar wavelength, and j is the imaginary unit with $j^2 = -1$. The Doppler frequency $f_{d,l} = 2v_{\mathrm{r}l}/\lambda$ induces the phase shift across the array in the l-th node.

In heterogeneous environments, the clutter $\mathbf{c}_l(k)$ is described by the product model: $\mathbf{c}_l(k) = \Delta_l(k)x_{0,l}(k)\mathbf{a}_{c,l}(k)$. Given the typically negligible internal motion of ground clutter, the spatial steering vector of the l-th radar is approximated as $\mathbf{a}_{c,l}(k) \approx [1, 1, \ldots, 1]^{\mathrm{T}}$. In the l-th node, the speckle component $x_{0,l}(k) \sim \mathcal{CN}(0, \sigma_c^2)$, while the texture variable $\Delta_l(k)$ that governs amplitude variations follows an inverse chi-square distribution, given by

$$f_{\Delta_l}(\delta) = \frac{2(\chi_l - 1)^{\chi_l}}{\Gamma(\chi_l)} \delta^{-(2\chi_l+1)} \exp\left(-\frac{\chi_l - 1}{\delta^2}\right), \tag{3}$$

where χ_l denotes the texture parameter of the l-th node indicating the degree of heterogeneity (smaller χ_l implies stronger heterogeneity).

III. Proposed Detection Method

The functional block diagram of the proposed detection framework is illustrated in Fig. 2. Briefly, the procedure commences by calculating the local detection test for each radar node in the distributed network, utilizing the γ-detection method described in [13]. Subsequently, these local tests are fused via a multiplicative rule to formulate the global detection test, expressed as $\Gamma = \prod_{l=1}^{L} \gamma_l$. To achieve a Constant False Alarm Rate (CFAR) detection, the statistical distribution of Γ is estimated to determine the decision threshold for a prescribed probability of false alarm (P_{fa}). The specific implementation steps are as follows.

Effective target detection in clutter-plus-noise environments necessitates effective clutter suppression. Without loss of generality, the adaptive matched filter is employed for the l-th radar node to optimally combine the array outputs. The clutter-suppression output of the l-th node is given by:

$$y_l(k) = \mathbf{u}_l^{\mathrm{H}}(k)\mathbf{z}_l(k), \tag{4}$$

where the optimal weight vector is defined as $\mathbf{u}_l(k) = (\mathbf{R}_l^{-1}(k)\mathbf{a}_{tl}(k))/(\mathbf{a}_{tl}^{\mathrm{H}}(k)\mathbf{R}_l^{-1}(k)\mathbf{a}_{tl}(k))$ and $[\cdot]^{\mathrm{H}}$ represents the complex conjugate transposition. For the l-th radar node, $\mathbf{R}_l(k)$ is the CCM, which can be estimated from Q independent training samples selected from the reference window surrounding pixel k:

$$\hat{\mathbf{R}}_l(k) = \frac{1}{Q} \sum_{q=1}^{Q} \tilde{\mathbf{z}}_{l,k}(q)\tilde{\mathbf{z}}_{l,k}^{\mathrm{H}}(q), \tag{5}$$

where $\tilde{\mathbf{z}}_{l,k}(q)$ represents the $M \times 1$ data vector of the q-th training sample. The magnitude-based statistic of the l-th node is defined as the normalized power of the residual:

$$\beta_l(k) = \frac{|y_l(k)|^2}{\sigma_{\mathrm{cn}l}^2}, \tag{6}$$

where $\sigma_{\mathrm{cn}l}^2$ represents the average power of the clutter residuals in the l-th radar node.

For the l-th radar node, two data vectors are constructed for a given pixel k from the SAR images corresponding to the first and last $M-1$ channels, as follows:

$$\mathbf{z}_{l1}(k) = [z_{l,1}(k), z_{l,2}(k), \ldots, z_{l,M-1}(k)]^{\mathrm{T}}, \tag{7}$$

$$\mathbf{z}_{l2}(k) = [z_{l,2}(k), z_{l,3}(k), \ldots, z_{l,M}(k)]^{\mathrm{T}}. \tag{8}$$

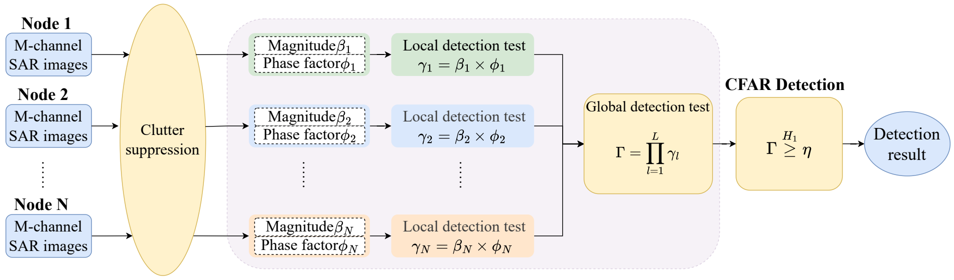

Fig. 2. Framework of the proposed detector.

$\mathbf{z}_{l1}(k)$ and $\mathbf{z}_{l2}(k)$ differ by a time delay of $d/(2v_{\text{p}l})$ caused by the time required for the platform to traverse the along-track channel spacing d at velocity $v_{\text{p}l}$. However, this delay does not impact the spatial steering vector of the l-th radar node, allowing the same optimal adaptive filtering weight vector $\mathbf{u}_{l1}^{\text{H}}(k)$ to be applied to both data vectors for clutter suppression. This process yields the residual signals of the l-th radar node:

$$y_{l1}(k) = \mathbf{u}_{l1}^{\text{H}}(k)\mathbf{z}_{l1}(k), \tag{9}$$

$$y_{l2}(k) = \mathbf{u}_{l1}^{\text{H}}(k)\mathbf{z}_{l2}(k). \tag{10}$$

where the weight vector $\mathbf{u}_{l1}(k)$ here is calculated in a manner analogous to $\mathbf{u}_l(k)$ described previously. Next, by applying complex interferometry to $y_{l1}(k)$ and $y_{l2}(k)$, the interferometric phase of the l-th radar node is extracted as:

$$\varphi_l(k) = \arg\left[y_{l1}(k)y_{l2}^{*}(k)\right], \tag{11}$$

where $[\cdot]^*$ donates the conjugate operation and $arg[\cdot]$ denotes the phase of a complex number within the 2π cycle. Under hypothesis H_1, the residual signals of the l-th radar node are given by:

$$y_{l1}(k) = y_{\text{s}1,l}(k) + y_{\text{c}1,l}(k) + y_{\text{n}1,l}(k), \tag{12}$$

$$\begin{aligned} y_{l2}(k) &= y_{\text{s}1,l}(k)\exp\left(j2\pi\frac{f_{tl}(k)d}{2v_{\text{p}l}}\right) \\ &+ y_{\text{c}1,l}(k)\exp\left(j2\pi\frac{f_{\text{c}l}(k)d}{2v_{\text{p}l}}\right) + y_{\text{n}2,l}(k), \end{aligned} \tag{13}$$

where $y_{\text{s}1,l}(k) = \mathbf{u}_{l1}^{\text{H}}(k)\mathbf{s}_{l1}(k)$, $y_{\text{c}1,l}(k) = \mathbf{u}_{l1}^{\text{H}}(k)\mathbf{c}_{l1}(k)$, $y_{\text{n}1,l}(k) = \mathbf{u}_{l1}^{\text{H}}(k)\mathbf{n}_{l1}(k)$, and $y_{\text{n}2,l}(k) = \mathbf{u}_{l1}^{\text{H}}(k)\mathbf{n}_{l2}(k)$.

When a moving target signal is present alongside clutter and noise, the clutter signal is almost completely suppressed, and the residual of the moving target typically exhibits a relatively large magnitude, i.e., $|y_{\text{s}1l}(k)| \gg |y_{\text{c}1,l}(k)| > |y_{\text{n}1,l}(k)|$. Consequently, the phase becomes $\varphi_l(k) \approx 2\pi\frac{f_{tl}(k)d}{2v_{\text{p}l}} \neq 0$. For the l-the node, the radial velocity of the moving target can be then estimated from the interferometric phase $\varphi_l(k)$ as:

$$\tilde{v}_{\text{r}l}(k) = \frac{\lambda v_{\text{p}l}}{2\pi d}\varphi_l(k). \tag{14}$$

Under hypothesis H_0, assuming a large stationary clutter residual is present, we have $|y_{\text{c}1,l}| \gg |y_{\text{n}1,l}| \approx |y_{\text{n}2,l}|$. In this case, the interferometric phase $\varphi_l(k)$ approximates to 0, since the clutter Doppler frequency $f_{cl}(k) \approx 0$.

Subsequently, a corresponding beamforming vector of the l-th radar node is constructed as follows:

$$\mathbf{d}_l(k) = \left[1, \exp\left(j2\pi\frac{\tilde{v}_{\text{r}l}(k)d}{\lambda v_{\text{p}l}}\right)\right]^{\text{T}} = \left[1, \exp(j\varphi_l(k))\right]^{\text{T}}. \tag{15}$$

Under hypothesis H_0, for a large clutter residual, $\tilde{v}_{\text{r}l} \approx 0$, and thus $\mathbf{d}_l(k)$ can be approximated as $\mathbf{d}_{l0} = [1, 1]^{\text{T}}$. For the l-th node, we proceed to estimate the subspace similarity between the residual's steering vector and this clutter basis vector:

$$\varepsilon_l(k) = \frac{|\mathbf{d}_l^{\text{H}}(k)\mathbf{d}_{l0}|}{\sqrt{|\mathbf{d}_l^{\text{H}}(k)\mathbf{d}_l(k)||\mathbf{d}_{l0}^{\text{H}}\mathbf{d}_{l0}|}}, \tag{16}$$

where $\varepsilon_l(k) \in [0, 1]$ is a function of the interferometric phase $\varphi_l(k)$. A larger value of $\varepsilon_l(k)$ indicates a higher degree of similarity between the residual and the clutter. For the l-th node, the local detection test is formulated by weighting the magnitude term with the phase factor $\phi_l(k) = 1 - \varepsilon_l(k)$:

$$\gamma_l(k) = \beta_l(k) \cdot \phi_l(k). \tag{17}$$

To leverage spatial diversity, the global test $\Gamma(k)$ is constructed via product fusion and immediately subjected to the binary hypothesis test:

$$\Gamma(k) = \prod_{l=1}^{L} \gamma_l(k) \underset{H_0}{\overset{H_1}{\gtrless}} \eta, \tag{18}$$

where the hypothesis H_1 is declared if $\Gamma(k)$ exceeds η; otherwise, H_0 is declared. η is the CFAR threshold satisfying $P_{\text{fa}} = \int_{\eta}^{\infty} f_{\Gamma|H_0}(x)dx$, where $f_{\Gamma|H_0}(x)$ is the PDF of Γ under H_0. The local tests from different distributed nodes are assumed mutually independent. According to the single-node analysis in [13], the PDF of each local statistic γ_l can be obtained individually. Since $\Gamma(k) = \prod_{l=1}^{L} \gamma_l(k)$, $f_{\Gamma|H_0}(x)$ follows from the product distribution of independent variables. Specifically, for two independent positive random variables $Z = \gamma_1\gamma_2$, the PDF $f_Z(x) = \int_0^{\infty} f_{\gamma_1}(t) f_{\gamma_2}\left(\frac{x}{t}\right)\frac{1}{t}dt,\ x > 0$, and the multi-node case follows recursively.

IV. Simulation Result

Based on the distributed signal model in Section II, simulation data are generated using the parameters in Table I. The clutter background comprises 300 homogeneous samples with a constant Clutter-to-Noise Ratio (CNR) of 10 dB and 700 heterogeneous samples with varying CNRs ranging from 15 to 50 dB, from which the degree of heterogeneity is estimated

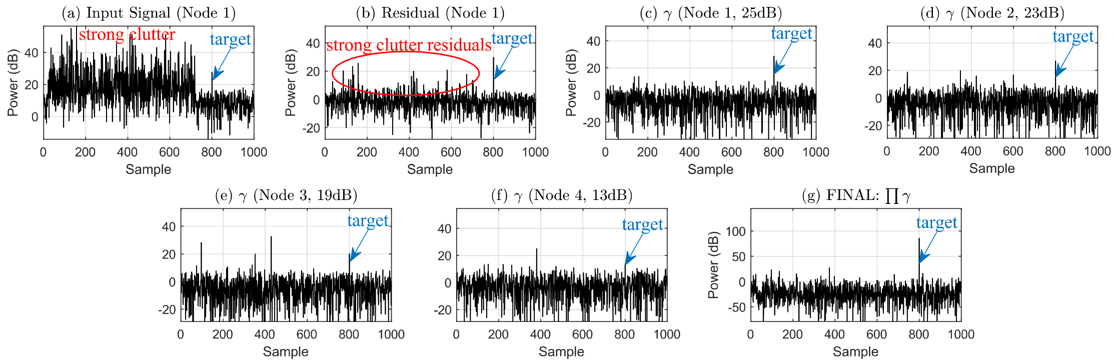

Fig. 3. Simulation results for the network: (a) magnitude of input SAR-image samples; (b) magnitude of the clutter-suppression residuals of Node 1; (c) γ-factor of Node 1; (d) γ-factor of Node 2; (e) γ-factor of Node 3; (f) γ-factor of Node 4; (g) Proposed

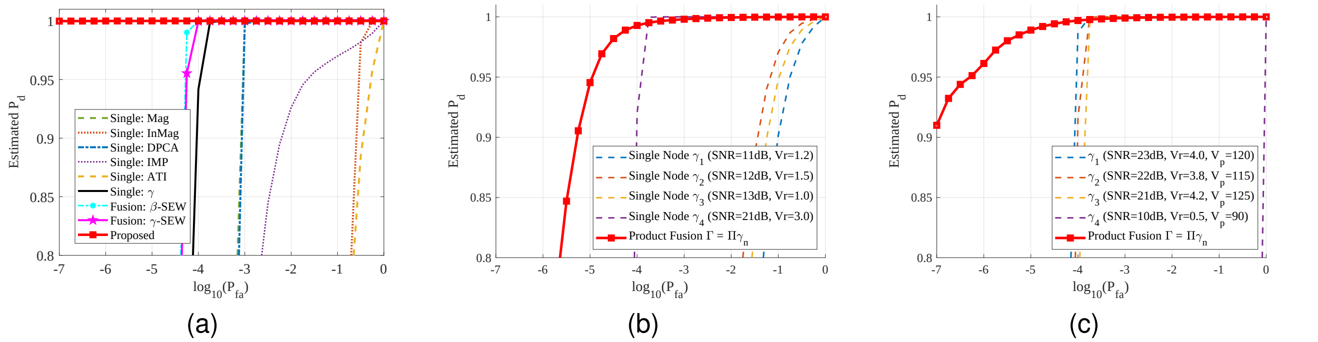

Fig. 4. Estimated ROC curves obtained from Monte Carlo simulations (10^5 trials). (a) Comparative detection performance against representative single-node detectors and distributed SEW fusion method. (b) Detection performance in a network with one superior node and three weaker nodes. (c) Detection performance in a network with the presence of a severely degraded node.

TABLE I
SIMULATION PARAMETERS FOR THE DISTRIBUTED RADAR NETWORK

Parameter	**Node 1**	**Node 2**	**Node 3**	**Node 4**
Wavelength (λ)	0.02 m			
Channels (M) / Spacing (d)	4 / 0.25 m			
Platform Velocity (v_p)	100 m/s	110 m/s	80 m/s	120 m/s
Target Radial Velocity (v_r)	2 m/s	1.8 m/s	1.2 m/s	1.6 m/s
Input SNR	25 dB	23 dB	19 dB	13 dB

to be $\hat{\chi} \approx 2.1$. A moving target is inserted at range cell 800. To simulate a low-SCNR masking scenario, a strong clutter spike (CNR=30 dB) is superimposed directly at the target position. To account for inconsistent channel responses, a random channel phase error with zero mean and a variance of $0.5°$ is introduced.

To evaluate the detection performance, we conduct a comparative analysis against representative detection methods, including the single-node Magnitude (Mag), Along-Track Interferometry (ATI) [2], Displaced Phase Center Antenna (DPCA) [3], and Interferometry Magnitude and Phase (IMP) [8] detectors, as well as the distributed SEW fusion method. Fig. 3 presents the detection results from a single simulation trial.As observed in Fig. 3(b), strong clutter residuals inevitably persist in the clutter-suppression magnitude output due to the heterogeneous clutter background. 3(c)-(f), the local γ factor detector further suppresses these strong residuals; however, its performance degrades under low SNR (Node 4, 12 dB) or small radial velocity (Node 3, $v_\mathrm{r} = 1.0$ m/s) conditions. In contrast, Fig. 3(g) demonstrates that the proposed method effectively eliminates the remaining clutter, making the true target stand out prominently against the clutter background.

The Receiver Operating Characteristic (ROC) curves via Monte Carlo simulations is shown in Fig. 4(a). The proposed product fusion method demonstrates a remarkable performance advantage over single-node detectors and distributed fusion strategies, including both magnitude-based SEW (β-SEW) and γ-based SEW (γ-SEW). Furthermore, Fig. 4(b) and (c) indicate that the proposed method remains effective when only part of the distributed nodes are degraded. Note that Fig. 4(c) represents the case where one node falls into the blind-speed region of the local detector, so that its local statistic is significantly weakened, but not reduced to exactly zero. If one local statistic approaches zero, the product rule may degrade the global detection performance; in practice, a small positive lower bound can be imposed before fusion.

V. CONCLUSION

This paper proposes a product fusion based moving target detection metric with distributed SAR systems. The proposed method uses the multiplicative operation to fuse local detection decisions derived from the γ test in [13] at each node for global moving target detection. By leveraging the spatial diversity in the distributed network, heterogeneous clutter is effectively suppressed and the contrast between moving targets and background is largely enhanced. Several simulation results demonstrate that the proposed method significantly improves moving target detection performance in heterogeneous environments. Considering that practical phase synchronization and calibration errors may affect interferometric phase accuracy, measured-data validation will be pursued in future work.

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